

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-301 — Analog Digital Communication

Time: Three hours

Maximum Marks: 70

*Instructions: Answer the questions in each section as instructed.***SECTION — A**

Long Answer questions. Answer any two questions:

 $2 \times 12 = 24$ Marks**Question 1.** Define amplitude modulation (AM) and describe it in detail with suitable diagrams.**Question 2.** Determine the following for DSB-SC:

- (a) Bandwidth [2]
- (b) Time domain representation [5]
- (c) Frequency spectrum waveform [3]
- (d) Advantages, disadvantages, and applications [2]

Question 3. For a modulation coefficient $m = 0.2$ and an unmodulated carrier power $P_c = 1000$ W, calculate:

- (a) the total sideband power,
- (b) upper and lower sideband power,
- (c) modulated carrier power, and
- (d) total transmitted power.

Question 4. State the principle of angle modulation. Derive phase deviation, modulation index, frequency deviation, and percent modulation.**SECTION — B**

Question 5. Attempt any six parts (Answer in about 200 words each): **$6 \times 6 = 36$ Marks**

- (a) Define sampling and express the Nyquist sampling rate.
 - (b) Define elementary signals and describe them using graphs and mathematical functions.
 - (c) Define a system. How do we check the linearity of a system? Give a suitable example with proof.
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- (d) What is the Fourier transform? Calculate the Fourier transform of the following signals:
- (i) $e^{-at}u(t)$ where $u(t)$ is the unit step signal and $a > 0$
 - (ii) $e^{-at}u(t) + e^{at}u(-t)$
- (e) If $x(t) = u(t)$ is the unit step signal, sketch the following:
- (i) $3x(t - 2)$
 - (ii) $5x(-2t + 1)$
- (f) Classify the different methods of pulse modulation techniques.
- (g) Discuss the generation of DSB-SC waves.
- (h) A 1000 kHz carrier is simultaneously modulated with 300 Hz, 800 Hz, and 2 kHz audio sine waves. Determine the frequencies present in the output.
- (i) Write the advantages of digital communication over analog communication.

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) Analog signals cannot be sent using digital techniques. (True / False)
- (b) Digitizing a signal can reduce distortion. (True / False)
- (c) Digitization removes noise and distortion from analog signals. (True / False)
- (d) A digital signal can be changed from a 1 to a 0 by noise. (True / False)
- (e) Unlike analog, digital communications is not band-limited. (True / False)
- (f) The amount of digital data that can be sent is limited by the channel capacity (C). (True / False)
- (g) What are causal and non-causal signals?
- (h) What is the bandwidth of an AM signal?
- (i) Write the full form of DSB-FC, DSB-SC, and SSB-SC.
- (j) Write the Nyquist criterion for sampling.